

HAS A RELATIONSHIP

Connection between two objects is called as relationship in Java. Relationship is of two types.

Has A Relationship: Connection between two objects exists as dependency between them. The two classes are called Dependent class and Another class.

Is A Relationship: Connection between two objects exists as Parent – Child relationship between classes. The two classes are called Parent/Super/Base class and Child/Sub/Derived class.

HAS A RELATIONSHIP:

The connection between two objects exists as dependency between them. The dependency between them is again of two levels.

Composition: The level of dependency exists in such a way that if one object does not exist, the other object cannot survive. It is achieved by a technique called as Early Instantiation.

Aggregation: The level of dependency exists in such a way that if one object does not exist, the other object can survive. It is achieved by a technique called as Lazy/Late Instantiation.

Classes in Has-A Relationship:

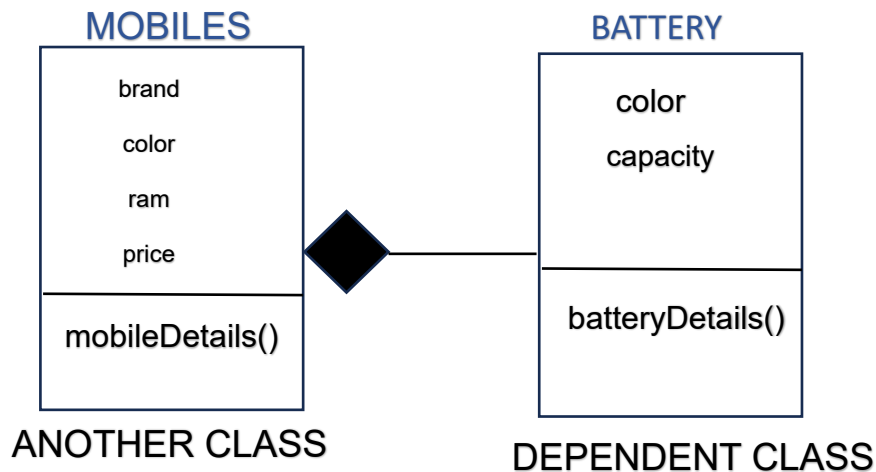
The classes are created for two real-world objects. One object is present inside another object.

DEPENDENT CLASS: The object which is smaller or present inside another object is called dependent class.

ANOTHER CLASS: The object which is bigger or holds another object is called another class.

COMPOSITION

Early Instantiation is the technique used in Composition.



Steps to achieve early instantiation:

1. Create Dependent class
2. Create Another class
3. Create Dependent class variable and place it in Another class
4. Create object for the Dependent class using Constructor of Another class

WHY CALLED AS EARLY INSTANTIATION:

The instantiation for the dependent class happens as in-line instantiation during another class instantiation. Hence, called as early instantiation.

```

class Battery
{
    String color;
    int capacity;
    Battery(String color, int capacity)
    {
        this.color = color;
        this.capacity = capacity;
    }
    public void batteryDetails()
    {
        System.out.println("*****BATTERY DETAILS*****");
        System.out.println("color of battery is: "+color);
        System.out.println("capacity of battery is: "+capacity);
    }
}

```

```

class Mobiles
{
    String brand;
    String color;
    int ram;
    float price;
    Battery bat; //place the dependent variable inside another class
    Mobiles(String brand, String color, int ram, float price, Battery bat)
    {
        this.brand = brand;
        this.color = color;
        this.ram = ram;
        this.price = price;
        this.bat = bat;
    }
    public void mobileDetails()
    {
        System.out.println("*****MOBILE DEATILS*****");
        System.out.println("brand of mobile: "+brand);
        System.out.println("color of mobile is: "+color);
        System.out.println("ram of mobile is: "+ram);
        System.out.println("price of mobile is: "+price);
    }
}

```

```

1 package javapackage;
2
3 class CompositionDriver
4 {
5     public static void main(String[] args)
6     {
7         Mobiles mob = new Mobiles("IPHONE","WHITE",8,125634.25f, new Battery("black", 1500));
8         //create the dependent class object using constructor
9         mob.mobileDetails();
10        mob.bat.batteryDetails();
11    }
12 }

```

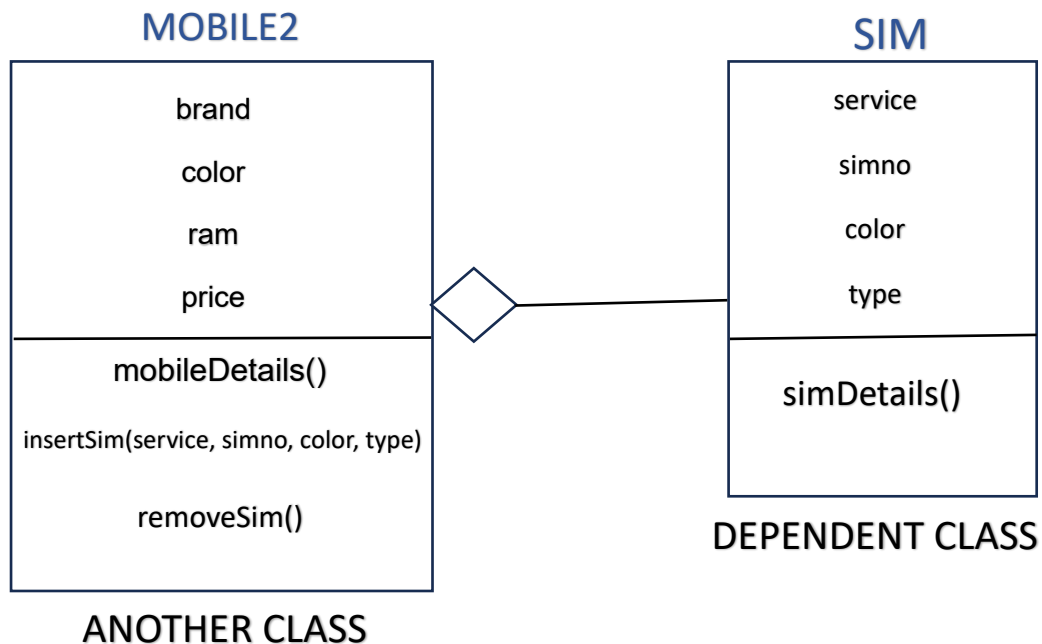
```

<terminated> CompositionDriver [Java Application] C
*****MOBILE DEATILS*****
brand of mobile: IPHONE
color of mobile is: WHITE
ram of mobile is: 8
price of mobile is: 125634.25
*****BATTERY DETAILS*****
color of battery is: black
capacity of battery is: 1500

```

AGGREGATION

Late/Lazy Instantiation is the technique used in Aggregation.



Steps to achieve Late/Lazy instantiation:

1. Create Dependent class
2. Create Another class
3. Create Dependent class variable in Another class
4. Create Helper method
5. Create Dependent class object in the helper method
6. Call the helper method whenever needed

WHY CALLED AS LATE/LAZY INSTANTIATION:

The instantiation for the dependent class happens separately in a method called helper method. Whenever an object for dependent class is needed, the helper method is called. Hence, it is called as Late/Lazy instantiation.

```

class Sim
{
    String service;
    int simno;
    String color;
    String type;
    Sim(String service, int simno, String color, String type)
    {
        this.service = service;
        this.simno = simno;
        this.color = color;
        this.type = type;
    }
    public void simDetails()
    {
        System.out.println("*****SIM DETAILS*****");
        System.out.println("service provider of sim is: "+service);
        System.out.println("number of sim is: "+simno);
        System.out.println("color of sim is: "+color);
        System.out.println("type of sim is: "+type);
    }
}

```

```

class Mobile2
{
    String brand;
    String color;
    int ram;
    float price;
    Sim slot; //place the dependent variable inside another class
    Mobile2(String brand, String color, int ram, float price)
    {
        this.brand = brand;
        this.color = color;
        this.ram = ram;
        this.price = price;
    }
    public void mobileDetails()
    {
        System.out.println("*****MOBILE DEATILS*****");
        System.out.println("brand of mobile: "+brand);
        System.out.println("color of mobile is: "+color);
        System.out.println("ram of mobile is: "+ram);
        System.out.println("price of mobile is: "+price);
    }
    public void insertSim(String service, int simno, String color, String type)
    {
        slot = new Sim(service, simno, color, type);
        //create dependent class object using helper method
        System.out.println("sim inserted successfully");
    }
    public void removeSim()
    {
        slot = null;
        System.out.println("sim removed successfully");
    }
}

```

```

1 package javapackage;
2
3 class AggregationDriver
4 {
5     public static void main(String[] args)
6     {
7         Mobile2 mob = new Mobile2("SAMSUNG","WHITE",16,65214.23f);
8         mob.mobileDetails();
9
10        mob.insertSim("jio", 52314, "brown", "postpaid"); //call helper method
11        mob.slot.simDetails();
12
13        mob.removeSim();
14
15        mob.insertSim("airtel", 60127, "green", "prepaid");
16        mob.slot.simDetails();
17    }
18 }
19
20
21
22
23

```

```

<terminated> AggregationDriver [Java Application] C:\Us
*****MOBILE DEATILS*****
brand of mobile: SAMSUNG
color of mobile is: WHITE
ram of mobile is: 16
price of mobile is: 65214.23
sim inserted successfully
*****SIM DETAILS*****
service provider of sim is: jio
number of sim is: 52314
color of sim is: brown
type of sim is: postpaid
sim removed successfully
sim inserted successfully
*****SIM DETAILS*****
service provider of sim is: airtel
number of sim is: 60127
color of sim is: green
type of sim is: prepaid

```